



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2024-25/EMLMIHT/1003510

Date:

03/03/2025(19:55:00)

Subject:Revision of LMI for Operation and Maintenance Guidelines for BCW Pumps

LMI Reference No:LMI/OGN/OPS/SYST/039

LMIs are reviewed periodically for further improvement and incorporation of recommendation by COS.

No changes envisaged by COS.

Review of LMI being done after two years.

Put up approval of competent authority please.

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checked and found in order

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NO Changes Envisag, Only Periodic Review

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As Entire Document is redrafted as per LMI Standard Format,It should be Rev.01 & to be mentioned on Every page instead of Rev. 0. Same is to be mentioned in record of revision.

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LMI/OGN/OPS/SYST/039 is prepared in line with CC OS Document.It may approved.

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LMI document has been reviewed and found inline with the OGN document.
forwarded for subsequent approval please.

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Reviewed and submitted for approval please.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

OPERATION AND MAINTENANCE GUIDELINES FOR BCW PUMPS

LMI/ OGN/OPS/SYST/039, REV:01

MAR - 2025

APPROVED FOR IMPLEMENTATION BY HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

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INDEX

SLNO	PARTICULARS	PAGE NO
1	INTRODUCTION	1
2	SUPERSEDED DOCUMENTS	1
3	DETAILS OF REVISION	1
4	GENERAL DESCRIPTION	1
5	OPERATIONAL ASPECTS	7
6	RECORDS OF PUMP STATUS	14
7	TRANSFER OF PUMP FROM ONE STATION TO OTHER	15
8	INSPECTION AND MAINTENANCE DETAILS	15
9	MAINTENANCE OF BCW PUMPS	16
10	CONDITION MONITORING AND CHECKS	17
11	PRESERVATION	18
12	SPARES MANAGEMENT	19
13	TROUBLESHOOTING	19
14	REPORTING	20
15	REFERENCES	20
16	REVIEW	20
17	CHECKLISTS	
	a. OPERATION DAILY CHECK LIST	21
	b. CHECKLIST- PRESERVATION AND INSPECTION CHECKLIST	22
	c. CHECKLIST- O&M CHECK LIST DURING START UP, EMERGENCY AND BLACK OUT CONDITION	23
18	ANNEXURES	
	ANNEXURE-1 : POPULATION OF BCW PUMPS AT NTPC SIMHADRI.	25
	ANNEXURE-2: PROCEDURE FOR WATER FILLED STORAGE OF A SPARE PUMP MOTOR SET HAYWARD TYLER PUMP	26
	ANNEXURE-3: REGULAR INSPECTION IN EVERY 3 MONTHS	28
	ANNEXURE-4: PROCEDURE FOR WATER FILLED STORAGE OF A SPARE PUMP MOTOR SET TORISHIMA PUMP	29
	ANNEXURE-5: POPULATION OF BCW PUMPS AT NTPC STATIONS & JV	31

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

1.0 INTRODUCTION:

Now a day's thermal power plants have been challenged to operate in flexible regime as well as maintain high reliability and availability. Reliability and availability of any power station depends on reliability of equipment.

BCW pumps play a vital role in ensuring reliability and availability of any power plant as outage of BCW pump can cause partial loss as well as forced outage of the unit. A good operation and maintenance practice can improve the reliability and availability of our equipment.

In NTPC Simhadri, BCW pumps of make Hayward Tyler in stage-1 units and Torishima Pump India Pvt Ltd are available in stage-2 units.

2.0 SUPERSEDED DOCUMENTS: LMI/OGN/OPS/SYST/039,Rev:0

3.0 RECORDS OF REVISION:

Rev-1: The LMI document is redrafted as per standard LMI format

Document reviewed on 27.08.2024 BY COS. No changes envisaged by COS.

4.0 GENERAL DESCRIPTION:

The BCW pump is a Glandless Motor Circulator Pump, which is designed, for recirculating boiler water through the boiler water systems of fossil fuel or liquid cooled nuclear reactor power stations and similar applications.

The circulators consist of a single stage centrifugal pump and a wet stator induction motor, which are mounted within a common pressure vessel. The vessel consists of three main parts, a pump casing, motor housing and motor cover.

The motor is suspended beneath the pump casing and is filled with cold boilerwater at full system pressure. No seal exists between the pump and motor, but provision is made to thermally isolate the pump from the motor in the following respect: -

- Thermal Conduction.** Because the pump temperature is so high, usually above 340°C and the motor temperature is limited to about 55°C, a simple restriction, in the form of a 'neck', is provided to minimize heat conduction.
- Hot Water Diffusion.** To minimize diffusion of boiler water, a narrow annulus surrounds the rotor shaft, between the hot and cold regions. A baffle ring restricts solids entering the annulus.
- Motor Cooling.** The motor cavity is maintained at a low temperature by a heat exchanger in a closed loop water circulation system, thus extracting the heat conducted from the pump.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

In addition, this water circulates through the stator and bearings, extracting the heat generated in the windings and providing bearing lubrication. An internal filter is incorporated in the circulating system.

d) **In Emergency conditions**, if low pressure coolant to the heat exchanger fails, or is inadequate to cope up with heat flow from the pump case, a cold purge can be applied to the bottom of the motor to limit the temperature rise generally in BCW pump. There are three sources of cold purge with different applicability.

1. Boiler fill line (Initial cold fill and purge)
2. CEP discharge (Temporary line during initial cold purge)
3. BFP Discharge (During acid cleaning or in emergency conditions)

NOTE: A cold purge is to be used only as an emergency

4.1 PUMP

The pump comprises a single suction and double (single in case of Super critical unit) discharge branch casing, welded into the boiler system pipework at the suction and discharge branches with the suction branch upper most. Within the pump cavity rotates a key driven, mixed flow type impeller, mounted on the end of the extended motor shaft. The impeller is hydraulically balanced to reduce axial thrust loads. 'Renewable' wear rings are fitted to both the front and back of the impeller. The impeller wear rings are the harder component to prevent galling.

4.2 MOTOR

The motor is a squirrel cage, wet stator, induction motor, the stator wound with a special watertight insulated cable. The phase joints and lead connections are also molded in an insulated material. The motor is joined to the pump casing by a pressure tight flange joint and a motor cover completes the pressure tight shell.

The motor shell contains all the moving parts, except for the impeller. Below the impeller is situated an integral heat baffle which reduces the heat flow, a combination of convection and conduction, down the unit. A baffle wear ring-cum-sleeve above the baffle forms a labyrinth with the underside of the impeller to limit sediment penetration into the motor. Should foreign matter manage to pass the labyrinth device into the motor enclosure, a filter located at the base of the cover end bearing housing strains it out.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

The motor design is such that for ease of maintenance, the stator shell, complete with the stator pack, the rotor shaft, the journal bearing housings and the thrust assembly, can be withdrawn from the motor case in sequence. Lifting lugs are supplied to secure hoists when raising and lowering the motor.

Insulation of the motor is generally XLPE polyethylene category of Y Class, Class-Y insulations can withstand a temperature of up to 90°C

4.3 Auxiliary Cooling Circuit (Fig. 1)

The motor is provided with its own auxiliary closed circuit, which besides cooling the motor lubricates the bearings. The water is continuously circulated through the bearings, motor windings and an external heat exchanger (cooler), by an auxiliary impeller at the thrust bearing end of the motor shaft.

When the motor is stationary, thermo syphon circulation takes place to remove conducted heat from the pump end of the motor.

4.4 Internal Filter

A stainless steel woven wire strainer, fitted at the base of the reverse thrust seat, filters the liquid in the motor before it is circulated through the bearings after passing through the heat exchanger (cooler).

The filter should be cleaned at normal maintenance periods, removing any accumulation of foreign matter in the motor cover.

4.5 Terminal Gland

The electrical supply to the motor is taken through special high-pressure single lead cable terminal glands of the high temperature, safety type.

The seal is affected by a terminal gland moldings, together with O-rings located between an insulating sleeve and an inner casing, and between the Inner casing and the motor case. The addition of a cone-shaped collar within the terminal moldings, plus the design of the inner casing, prevents any possibility of a gland blowout at high temperatures. However, if the motor is subjected to sustained overheating, a slight leakage may occur into the terminal box.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

4.6 TERMINAL BOX

Robust fabricated steel terminal boxes are provided on the side of the motor casing for connecting the power supply to the motor.

The power supply cable for the motor enters the box at the bottom and must be sealed by a cable gland. Each phase notation is identified by the coding, Red, Yellow or Blue and IEC markings 'U', 'V' & 'W'. See Section 4 – Installation.

Bursting discs are fitted to the wall of the terminal box to relieve excessive pressure build up within the box should a flash over occur at the terminals. Desiccators are also fitted to absorb atmospheric moisture that may increase the electrical resistance to ground. The whole assembly is completely enclosed by a bolted on cover.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

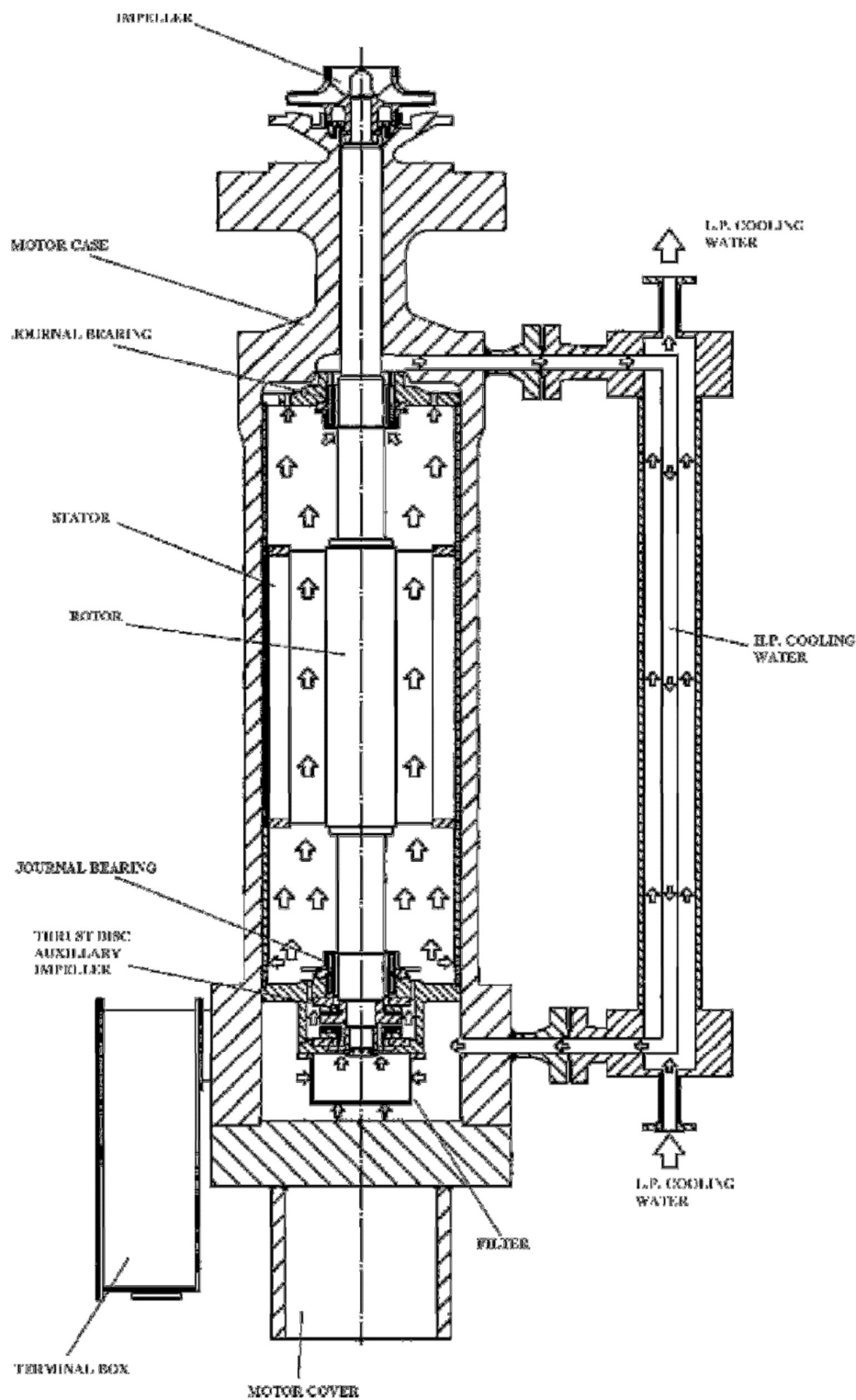


Fig. 1

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

4.7 HEAT EXCHANGER

A heat exchanger (cooler) is fitted to dissipate the heat generated by the motor and the heat transferred from the pump casing. Brackets are provided on the motor case to mount the heat exchanger. The heat exchanger is of the shell and tube type comprising two main components; the pressure vessel which houses the tube bundles, and the inlet and outlet covers. The high-pressure water from the motor flows through the tubes whilst the low-pressure cooling water flows through the shell. High pressure outlet and inlet flange connections are situated top and bottom of the motor case for connection to the high-pressure mating flanges.

4.8 HEAT BARRIER

In some pumps provided by KSB and Torishima, there is a heat barrier between the pump and motor parts. Its purpose is to separate the hot pump part from the cold motor part and to reduce the heat transfer to a minimum.

This heat barrier is supplied with low pressure cooling water similar to the heat exchanger discussed earlier.

4.9 ALARMSYSTEMS

a. Thermocouples

Three thermocouples are provided to monitor the motor case cavity temperature. An alarm circuit is provided such that when motor temperature reaches a predefined value as prescribed by OEM alarm appears and a subsequent trip signal is provided if temperature reaches the trip set point as prescribed by OEM.

Two further thermocouples are provided to monitor the pump case metal temperature.

b. Dial Thermometers

Temperature indicating dial thermometers are provided on the motor case, on the low pressure heat exchanger pipe work assembly (inlet and outlet). These give local visual indication only.

c. Flow / Switch Indicator

A flow/switch indicator is supplied for connecting into the low pressure cooling water circuit (inlet) of the heat exchanger. An alarm circuit may be arranged to indicate low flow below the set point.

d. Flow Sight

A flow sight is provided for the LP cooling water circuit (outlet).

4.10 STRAINER

A 'Y' type strainer is provided to prevent large particles from entering the motor fill line. It consists of a Y type body with a bolted cover and gasket, containing a 60-mesh stainless steel filter screen.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

5.0 OPERATIONAL ASPECTS (Operation. Scope)

5.1 FILLING AND PURGING OF BCW PUMPS.

There are basically three sources for BCW cavity purging which are:

1. Boiler fill line (LP fill and purge line)
2. CEP discharge* (Temporary connection)
3. BFP discharge* (HP fill and purge line)

(*not available at Super critical units)

During initial filling and purging only LP fill and purge line from boiler fill line is to be used. Filling the circulator correctly is a prerequisite to ensure satisfactory operation. If air is present in the motor it could affect bearing lubrication and precipitate bearing failure. If air pockets are present in the stator this would affect dissipation of heat generated in the windings, creating hot spots and consequent degrading of the insulating material, and ultimately cause winding failure.

Due to the complexity of the passages in the motor through which water must flow to displace all air, it is necessary to fill the motor very slowly. The motor must NOT be filled via the pump casing, but only through the filling connector located at the motor cover in bottom.

- 1) Isolate the circulator from the boiler by closing appropriate discharge valves, and drain valves.
- 2) Flush out the piping for the supply of low pressure treated boiler water from boiler fill pump, for filling the motor, until clean air free water is discharged and chemistry parameters are met, through the blow down valve just before the BCW motor cavity inlet line.

NOTE: The flushing must always be carried out before introducing any water through the lines to the motor and heat exchanger till the conductivity of water is achieved below **1 micro siemens/cm**.

- 3) Adjust the **flow rate to approximately 2 liters/minute** (or as specified by OEM)
- 4) After connecting the supply, open the heat exchanger vent on the high pressure side, fill the motor from the bottom and vent air from the top, until clean air free water is discharged from the vents.
- 5) Close the vent on the high pressure side of the heat exchanger.
- 6) Continue to fill the motor and vent any air via the vent connections in the boiler system pipe work till water comes through it with the same chemistry parameters as inlet to BCW pump.
- 7) Isolate the now filled circulator.
- 8) Ensure that the terminal box is dry and with the unit full of water below 20°C or at normal ambient temperature, using a **1,000 Volt Megger, check winding resistance to ground at the terminals. The resistance should**

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

exceed 200 mega ohms.

IF NOT, CALL OEM. DO NOT ATTEMPT TO RUN THE MOTOR.

Note: IR measurement should also be done before filling the motor cavity for analyzing and comparing the values after cavity filling.

Responsibility: UCE

5.2 PRE-START CHECK

The following pre-start checks must be satisfied before starting the motor.

- 1) Ensure that the circulator is completely filled and vented with treated water as described in 5.1.
- 2) Ensure that the low pressure cooling water (DMCW) valves are open and the cooling water flow rate agrees with that specified in the heat exchanger manufacturer's specification. All other valves must be closed.
- 3) Ensure that the high pressure cold water purge is connected but isolated.
- 4) Using a 1000V megger, check that the insulation resistance to ground of the motor leads conforms to the Winding Resistance curve provided by OEM when measured at the terminal stems with the unit full of water at normal ambient temperature.
- 5) Check that the correct voltage is connected to the circulator.
- 6) Check that the electrical supply is available at the point of control selected and that the starting equipment is functioning correctly.
- 7) Test the operation of all instrumentation and alarms.
- 8) Ensure that sufficient NPSH is available (adequate level in Drum / Separator) for the pump to run without cavitation by ensuring adequate water level in drum/Separator and pump line-up.

Responsibility: UCE

5.3 INITIAL STARTING-COLD BOILER CONDITION

When a boiler is started from cold, a minimum of two circulators (where 3 BCW pumps are there) be operated. It is normal practice to simultaneously warm the standby circulators to permit immediate starting when required.

CAUTION:

- A. **Motors must not be energized if the motor cavity temperature is below 2°C.**
 - B. **There should be at least ten-minute intervals between two repetitive starts. This must never be exceeded; otherwise the motor winding temperature will rise and could damage the winding insulation. The motor may be damaged if motor temperature exceeds 65°C.**
1. Ensure that the pre-start checklist, as described earlier in section 4.3 has

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

been satisfied.

2. Make sure that the boiler is full and open the discharge valves.
3. Switch on power to the starter.
4. Press the starter button on the control console and energize the motor for 5 seconds only.

CAUTION: The motor runs up to full R.P.M. in appropriately one second. If the motor fails to start after five seconds, press the stop button and do not attempt to restart the circulator for 20 minutes.

5. After ten minutes pause, run the circulator for a further (2nd) five seconds. During this run, check the following: -
 - a) Motor Current (after ammeter has come off initial surge of current)
 - b) Differential pressure - The differential pressure should rise immediately to approximately 3 bars (or as specified by OEM).

NOTE: If the differential pressure does not rise, stop the motor immediately. It may be an indication that the motor is running in reverse. Reverse rotation will result in approximately 30% less head generated and about 10% more power absorbed than specified.

6. After another (3rd) ten-minute pause, energize the motor and run for twenty minutes. During this running period, carry out the following checks: -
 - a) Check the motor for vibration, using a vibration detector, several times and record reading for future comparison. Check for rubbing or excess bearing noise by holding a listening rod against the pump and motor case.
 - b) Check the motor operating temperature at regular intervals. Initially it should rise several degrees then stabilize. If necessary, adjust the secondary cooling water flow after checking that the strainers are not obstructed.
 - c) Check, as the motor warms, that the high-pressure circulation pipe entering the heat exchanger from the top of the motor becomes warmer than the high pressure circulation pipe from the bottom of the heat exchanger to the motor case. This indicates correct functioning of the heat exchanger.
 - d) Check that motor current and differential pressure readings and record the readings.

Note: In Subcritical boiler with 3 BCW pumps, Normally two BCW pumps are kept in service, the availability of third BCW pump should be ensured by keeping any casing warming and circulation system in service.

Responsibility: UCE

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

5.4 LOW PRESSURE COOLING WATER SUPPLY FAILURE.

If the cooling water to the heat exchanger is lost while the circulator is at operating temperature, the motor will be de-energized as soon as the motor temperature reaches the trip setting of 65°C by the temperature controller. Prior to this an alarm will be initiated by the low pressure cooling water flow monitor indicating that the low pressure cooling water supply has been lost and a further alarm will be initiated by the temperature controller when the motor temperature reaches 60°C.

NTPC Simhadri Stage # 1 and Stage # 2 BCW Pumps has emergency cooling water circuit which consists of:

- 5.4.1 Overhead Tank (Fig-3)
- 5.4.2 Emergency Cooling supply valve (generally pneumatic On-Off type) (Fig-2)
- 5.4.3 Emergency cooling drain valve (generally pneumatic On-Off type) (Fig-2)
- 5.4.4 Two No. Booster p/p for overhead tank make-up (Fig-3)

In case of black out or low pressure cooling water supply failure to BCW pumps, water to BCW pumps is supplied via overhead tank by auto opening of emergency water supply valve to CW inlet manifold and draining the water from CW return manifold by auto opening the drain valve as shown in Fig-2.

Note: In case of failure of opening of emergency valves in case of blackout or LP cooling water failure, these valves should be opened manually

The emergency overhead tank is filled with the help of two booster pumps which operates on auto logic based on emergency tank level and takes suction from DMCW discharge header.

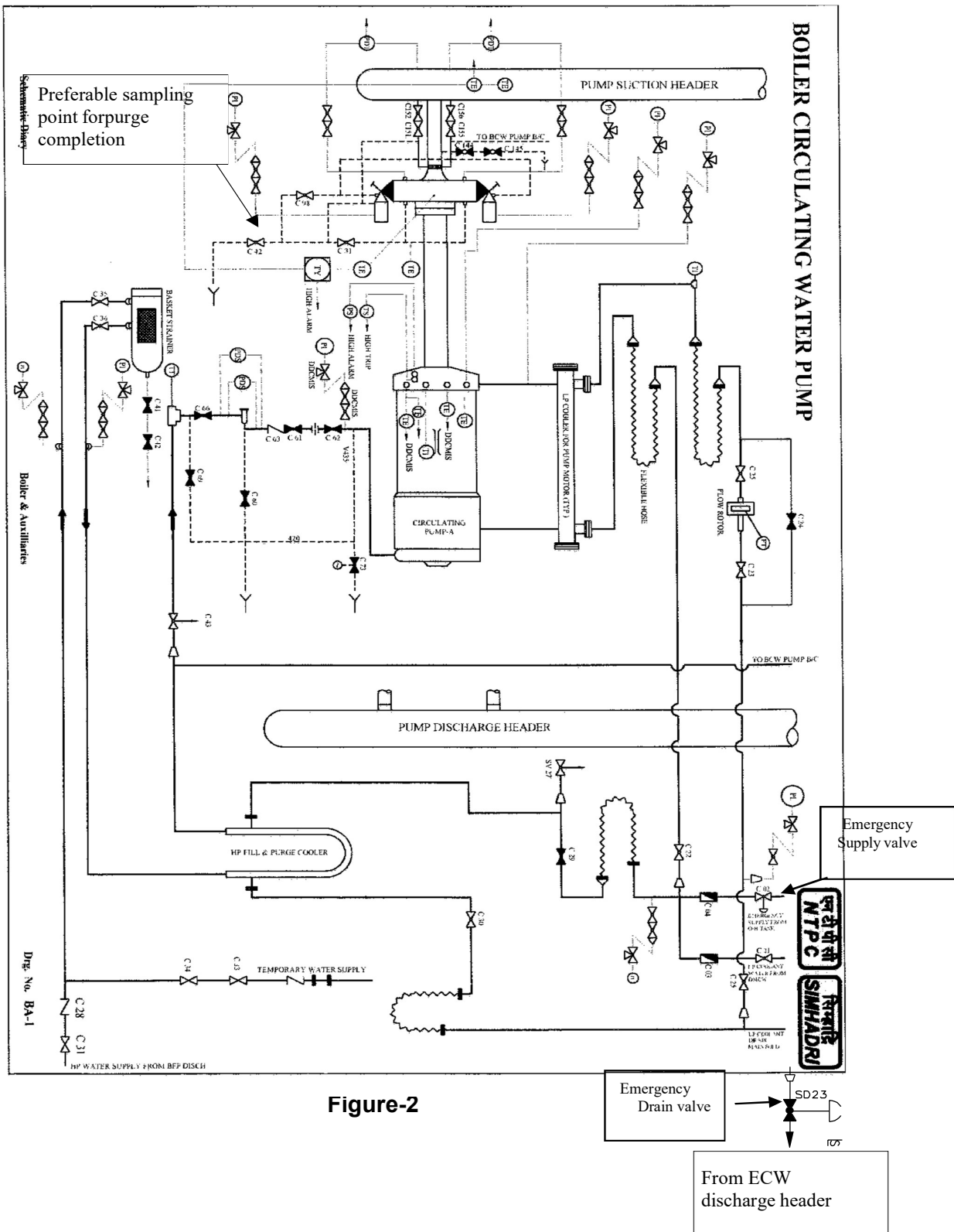
CAUTION: Damage to the windings may occur if the motor remains in operation after 5 minutes from the loss of cooling water.

CAUTION: Do not attempt to operate the motor by overriding the temperature controller trip mechanism and restore the low pressure cooling water as quickly as possible. Damage to the windings may occur if the cooling water is not restored before the motor temperature rises above 65°C.

NOTE: Before re-starting a circulator after an emergency shutdown, low pressure cooling water must be supplied to the pump to reduce the motor temperature to at least 38°C.

Responsibility: UCE

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1



TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

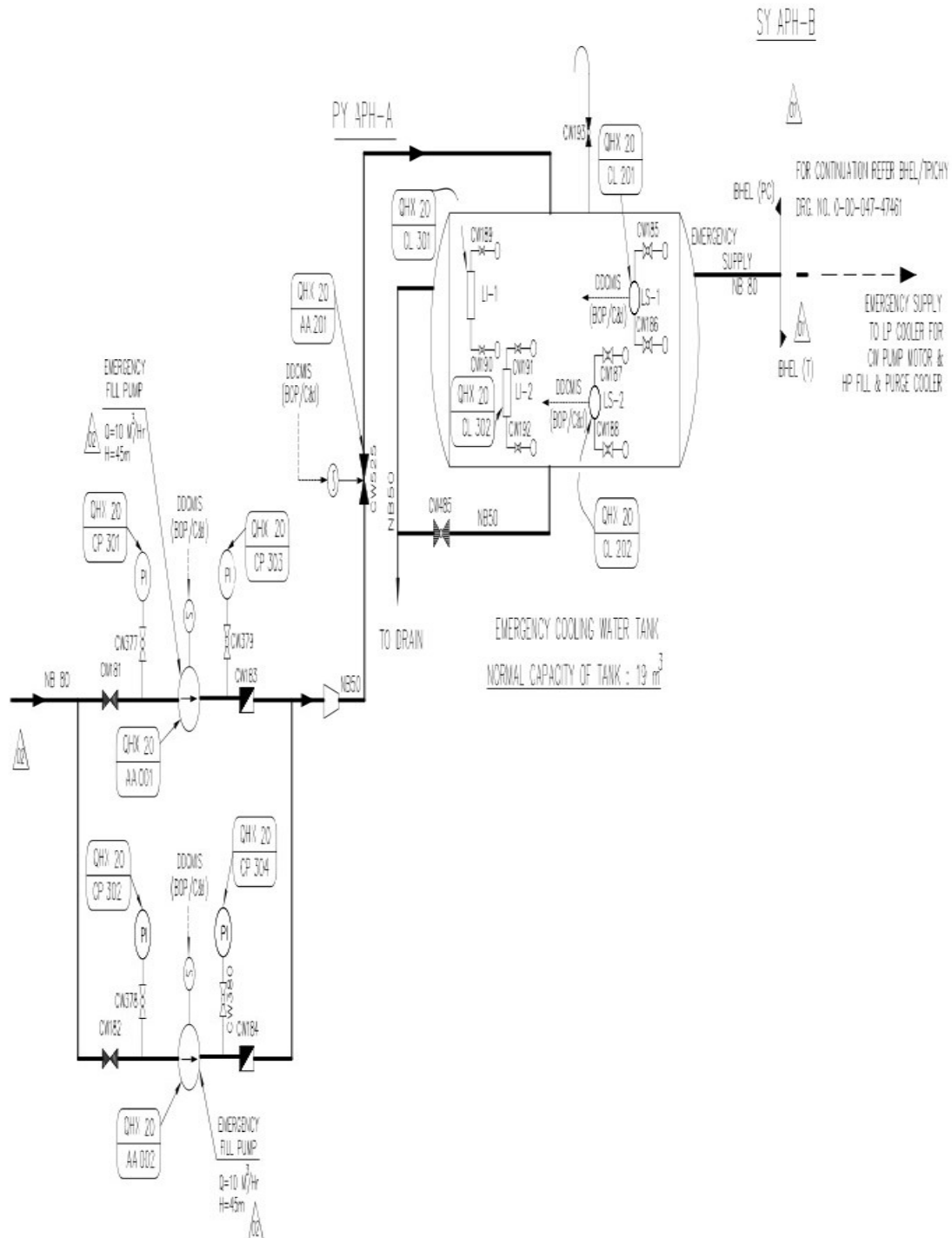


Figure-3

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

5.5 BOILER INITIAL CLEANING AND OTHER EMERGENCIES

5.5.1 During boiling-out or acid cleaning operation of the boiler, the de-energized circulators must be isolated completely prior to introducing any chemicals to the boiler. The circulators must be continuously purged with clean cool water at a pressure of approximately 7 bar, in excess of the existing drum pressure to eliminate infiltration of contaminated water into the bearing and motor components. This purging shall be done by HP fill and purge line from BFP, which consists of cooler and filter in the circuit as shown in Fig.2.

5.5.2 The HP fill and purge line can be charged in emergency conditions also if BCW (Running or stand-by) cavity temperature is rising beyond the permissible limit. Feed water conductivity may be checked before charging it to the BCP.

Note: Any leakage in motor cavity water in motor side and heat exchanger side should be taken very seriously as it can cause damage to motor. Unit must be stopped to attend the same and all actions must be taken to prevent the cavity temperature rising beyond the permissible value.

Note: Any passing in the fill and purge line can also cause the loss of cavity water and rise in cavity temperature, proper closing of fill and purge valve should be ensured before putting the circulator in service to prevent any backpassing.

Responsibility: UCE

5.6 HYDROTESTING BOILER WITH CIRCULATOR INSTALLED

Treated Water or Condensate Available: -

The hydrostatic test pressure must not exceed the maximum pressure specified by OEM.

Treated Water Not Available: -

If the boiler is to be hydrotested at an early state of construction, the pump casing can be pressure tested without the motor once the blank off plate is installed.

Post Hydrostatic Test Procedure

Boiler circulators are not to be drained but left filled to a level above the suction down comer.

Responsibility: UCE

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

5.7 STOPPING

- 1) Press the stop button on the control console. The run-down time is approximately 2 seconds. Close the discharge valves before draining the boiler.
- 2) Maintain the low-pressure cooling water supply to the heat exchanger till complete boiler cools down to normal temperature.

Responsibility: UCE

NOTE : On hot boilers, and whenever the circulator is run, the low pressure cooling water to the heat exchanger must always be on.

WARNING: Never drain the pump casing through the motor fill & drain line as this could cause contamination of the motor.

5.8 EXTENDED SHUT-DOWN

When the circulator remains shutdown for an extended period: -

- 1) Check regularly that it remains full of water.
- 2) **Run the circulator for a minimum of ten minutes every month or as prescribed by OEM.**

Responsibility: UCE

6.0 RECORDS OF PUMP STATUS

Information Database to be kept at stations.

- a) Pump type i.e. Name of Manufacturer, Type, and Serial no.
- b) Operating limits
- c) Specified life, number of starts and duty running hours.
- d) General arrangement, detail drawings and assembly details e.g. fits and clearances of pump assembly, support structure and associated pipework of sufficient detail for all overhaul and repair purpose.
- e) Technical data for each pump assembly in sufficient detail for performance investigations and operation monitoring, including the minimum net positive suction head required.

Responsibility: Operation

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

7.0 TRANSFER OF PUMP FROM ONE STATION TO OTHER

When a pump leaves site for storage or installation elsewhere, the Station General Manager should transfer a copy of the database to the receiving location.

8.0 INSPECTION AND MAINTENANCE DETAILS

Details of Records regarding the maintenance of pumps should be kept as follows as per annexure 2

- 8.1.1 Date of Installation
- 8.1.2 Dates of overhauls/inspections
- 8.1.3 Last Overhauling carried information.
- 8.1.4 Summaries of findings in each case including:
 - 8.1.4.1 Components or locations inspected
 - 8.1.4.2 Method of inspection
 - 8.1.4.3 Results e.g. clearance and NDT data (DPT of Thrust Pad etc)
 - 8.1.4.4 Action taken
 - 8.1.4.5 Components replaced or relocated with dates
 - 8.1.4.6 Date of next planned overhaul

Responsibility: EMD

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

9.0 MAINTENANCE OF BCW PUMPS

Maintenance of BCW pump should be done to

- a) To identify the content and timing of maintenance activities
- b) To outline a procedure for formulating repair/refurbishment

schemes. Maintenance Activities should be done as follows:

- A. Maintenance activities should be carried out in line with the original Equipment manufacturer's recommendations and to the quality assurance standards applicable to pump.
- B. In addition, the results of previous inspections and problems identified during operation & in similar plant elsewhere should also be taken into account for maintenance activities (Refer Annexure- 3 for other stations BCW details).
- C. Regarding Any Motor removed after a defect, Necessary preparation to be made for dispatch Motor to OEM Works within 30 Days or repair shall be carried out as early as possible in presence of OEM.
- D. Any mechanical welding, work on pumps, associated piping works & its alignment shall be done by Certified welder under supervision of Boiler maintenance Personnel.
- E. Approach's, staircase / platform required for carrying safe maintenance shall be made available at site (BMD Scope) .

Responsibility: EMD&BMD

9.1 MAJOR INSPECTION

Major / Complete inspection shall be carried out after every 6 years in as per recommendation of OEM.

The criteria for replacement, e.g. too large bearing clearances or clearance gaps, scratched contact surfaces (seizing), etc. If the measured clearance gaps or bearing clearances are close to the limit or if they exceed the maximum permissible size, new wear parts have to be fitted.

Check cooler for the presence of dirt on the low-pressure side and clean, if required. Check the cooler for leaks on the high-pressure side. For Test pressure refer the manual

The major inspection/ OH of BCW pumps should be carried out at site under the supervision of OEM

Responsibility: EMD

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

9.2 MINOR INSPECTION

Minor Inspection shall be carried out after every 2 years along with the boiler overhauling.

- Check the suction Strainer for any dirt
- Bottom Thrust pad inspection
- Bearing
- Flushing of the Coolers on both HP water and LP water sides

Responsibility:EMD

10.0 CONDITION MONITORING AND CHECKS

10.1 Daily Checks (Operation scope checklist to be prepared)

- Pump Suction & Diff pressure
- Motor Current
- Motor Cavity HP cooling Water Temperature
- Motor Cooler LP Cooling water outlet temperature
- Heat Barrier LP cooling water outlet
- Pump abnormal noise
- HP cooler water line leak
- LP cooler water line leak

Responsibility: UCE

10.2 Monthly checks (Operation scope checklist to be prepared)

- Motor insulation resistance (As per availability). monthly for RSD units.
- BCW pump, motor and associated piping should be free to expand, there should not be any resistance/ hindrance for the expansion.
- Standby pump start up checks
- **Note:** Schedule change over should be carried out in BCW pumps on fortnightly basis.

Responsibility: UCE

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

10.3 Opportunity Checks

- All BCW P&I checks to be done on every opportunity.
- The availability of ECW booster pumps (overhead tank make-up) should be ensured by taking a weekly trial as per the changeover/ trial schedule.
- Availability of emergency low pressure cooling water circuit to BCW pumps also needs to be ensured on every opportunity or once in three months by operating the emergency water supply and drain valve by simulating the cooling water supply fail signal.
- Adequacy of emergency overhead tank level also needs to be ensured on daily basis and logged in the operator log. (Alarm for emergency overheattank level hi should always persist)

Responsibility: UCE

11.0 PRESERVATION

a) Preservation for Spare BCW pumps

All BCW pumps Spare motors should be preserved in line with the procedure as mentioned in Annexure-2.

Responsibility: EMD

b) Preservation of BCW pumps during boiler preservation

(Responsibility: OPN)

During dry preservation of boiler, the following action with respect to the BCW pumps may be taken as per the OEM recommendation (Torishima Pumps, Hayward Tyler)

- Motor cavity water should not be drained during RSD/ Extended Shut- Down.
- Any water leakage from Cavity shall be topped up as per procedure OEM Manual.
- Before restart cavity water to be replaced by Purge, water as per OEM Manual.
- Ensure that the terminal boxes are dry
- Before start of the Motor, the IR value of the motor is to be checked (with cavity filled with Water) and the value should be more than equal to 200 MΩ at 1.0 kV.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

12.0 SPARES MANAGEMENT

- a) All spare parts not needed immediately are to be stored in a suitable place where they will not be impaired by contamination, humidity or other harmful influences.
- b) Keep termination kits away from chemical substances, solvents, fuels, acids, etc. Also, keep them protected from the light. This applies, in particular, to direct sunlight or strong artificial light high in ultraviolet rays.
- c) Quarterly inspection of BCW spares by EMD preservation committee

Responsibility: EMD

12.1 QUANTITY OF SPARES (As per EMD spares register and SAP Quantity)

- a) Minimum 1 spare Motor & pump set shall be available for every 3 Motors.
- b) As wear rates for replaceable items are dependent on plant conditions, type of duty, frequency of operation etc. Recommended spares list from the OEM shall be obtained. Minimum 2 sets of spare kits & consumables shall be available at site.

Note: The custodian of Electrical equipment like motors and etc. is EMD, however pumps and other boiler fittings (mechanical) regarding BCW pumps shall be maintained by BMD.

Responsibility: EMD &BMD

13.0 TROUBLESHOOTING (As per Trouble shooting checklist)

1. Increase in the required drive power: During normal pump operation, the current consumption should not exceed the nominal current by more than 10%. Carry out the following checks after the motor is isolated
 - Compare the specified duty data with the actual values.
 - Check the strainer in the suction line.
 - Check whether the pump is running in the overload range as a result of a change in the boiler resistances.
 - Check the bearings (radial bearing/thrust bearing) for wear.
 - Check whether the impeller seizes as a result of shaft unbalance or unsteady /eccentric rotation or any other Mechanical locking, if so overhauling should be done.
2. Sudden increase in the motor temperature
 - Check the low--pressure cooling water system
 - Cooling water flow monitoring device operating, i.e. the flow is at

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

least 70% of the nominal rate of flow.

- Check the temperature of the cooling water
- Check the low pressure system for leaks
- Check the flange joints between pump and high-pressure cooler for leaks.

3. Abnormal noise and vibrations

- Bearing malfunction: Overhaul and exchange parts unless other causes are apparent
- Scorching due to foreign matter ingress
- Check the points below if the circulating pump is found to produce abnormal noise and vibrations.
- Is the position of the gate and other valves in the system correct?
- Have the pipelines been connected correctly, free of tension or stress? Have the suspensions been mounted correctly?

Responsibility: Operation

14.0 REPORTING

ORT report shall cover all the abnormal operation / Defects Like also

- Any tripping or non-availability of pump
- Stations running with more than the designed no of pumps (all 3 pumps running in 500MW)
- Status of Repair of defective pump
- Spares availability status along with Consumables Kit

Responsibility: MTP

15.0 REFERENCES

- COS-LMI/OGN/OPS/SYST/039,Rev:0

16.0 REVIEW

Document shall be reviewed once in every two years by HOP (Simhadri).

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

17.0 CHECKLISTS

a. Operation Daily Check list

S.no	Parameter	Responsibility	Frequency
1	Pump Suction & Diff pressure	Operation	Daily
2	Motor Current	Operation	Daily
3	Motor Cavity HP cooling water temperature	Operation	Daily
4	Motor Cooler LP Cooling water outlet temperature	Operation	Daily
5	Heat Barrier LP cooling water outlet	Operation	Daily
6	Pump Abnormal noise	Operation	Daily
7	HP cooler water line leak	Operation	Daily
8	LP cooler water line leak	Operation	Daily
9	Run the circulator for a minimum of ten minutes every month (Not in operation more than one Month)	Operation	Monthly

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

b. Preservation and Inspection checklist

S.no	Parameter	Responsibility	Frequency
1	Run the circulator for a minimum often minutes every month	Operation	Monthly (When not in operation more than a month)
2	Approach's, staircase / platform required for carrying safe maintenance shall be made available at site.	BMD	Once
3	Major / Complete inspection shall be carried out after every 4- 5 years in as per recommendation of OEM.	EMD	4 – 5 Year
4	Quarterly inspection of BCW spares by EMD preservation committee	EMD/BMD	Quarterly
5	Minor Inspection shall be carried out after every 2 years along with the boiler overhauling. a) Check the suction Strainer for any dirt b) Bottom Thrust pad inspection c) Bearing d) Flushing of the Coolers on both HP water and LP water sides	EMD (In presence of OEM)	2 – Year at the boiler
6	Before start of the Motor, the IR value of the motor is to be checked (with cavity filled with Water) and the value should be more than equal to 200 MΩ at 1.0 kV.	EMD & Operation	Monthly for RSD units
7	Motor insulation resistance (As per availability).	EMD & OPN	Monthly for RSD units
8	All BCW pumps Spare motors should be preserved in line with the procedure given by OEM	EMD	Quarterly

TITLE	Operation and Maintenance Guidelines for BCW Pump
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

c. O&M CHECK LIST DURING START UP, EMERGENCY AND BLACKOUT CONDITION

Sr. No.	Description	Limiting Value	Type of Checks	Responsibility
1	A cold purge is to be used only as an emergency of BCW Pumps		Emergency	Operation
2	Motor and heat exchanger water conductivity of water is achieved below 1 micro siemens/cm	1 micro siemens/cm	OH/Motor replacement	Operation
3	When a boiler is started from cold, a minimum of two circulators (where 3 BCW pumps are there) be operated. It is normal practice to simultaneously warm the standby circulators to permit immediate starting when required.		Initial starting-Cold Boiler	Operation
4	Motors must not be energized if the motor cavity temperature is below 2°C.		Start of BCW Motor	Operation
5	There should be at least ten-minute intervals between two repetitive starts			Operation
6	The motor may be damaged if motor temperature exceeds 65°C.			Operation
7	The motor runs up to full R.P.M. in appropriately one second. If the motor fails to start after five seconds, press the stop button and do not attempt to restart the circulator for 20 minutes.			Operation
8	If the differential pressure does not rise, stop the motor immediately; it may be an indication that the motor is running in reverse. Reverse rotation will result in approximately 30% less head generated and about 10% more power absorbed than specified.			Operation

TITLE	Operation and Maintenance Guidelines for BCW Pump
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

9	A) In case of failure of opening of emergency valves in case of blackout or LP cooling water failure, these valves should be opened manually B) Damage to the windings may occur if the motor remains in operation after 5 minutes from the loss of cooling water C) Before re-starting a circulator after an emergency shutdown, low pressure cooling water must be supplied to the pump to reduce the motor temperature to at least 38°C.	Motor temperature should be less than 65 Degree	During Black out	Operation
10	Any leakage in motor cavity water in motor side and heat exchanger side should be taken very seriously as it can cause damage to motor		Monthly	Operation
11	Before start of the Motor, the IR value of the motor is to be checked (with cavity filled with Water) and the value should be more than equal to 200 MΩ at 1.0 kV	200 M Ohm with 1.0 KV Insulation Tester		EMD & Operation
12	On hot boilers, and whenever the circulator is run, the low pressure cooling water to the heat exchanger must always be on.			Operation
13	Never drain the pump casing through the motor fill & drain line as this could cause contamination of the motor.			Operation

TITLE	Operation and Maintenance Guidelines for BCW Pump
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

Annexure-1**Population of BCW pumps at NTPC Simhadri.**

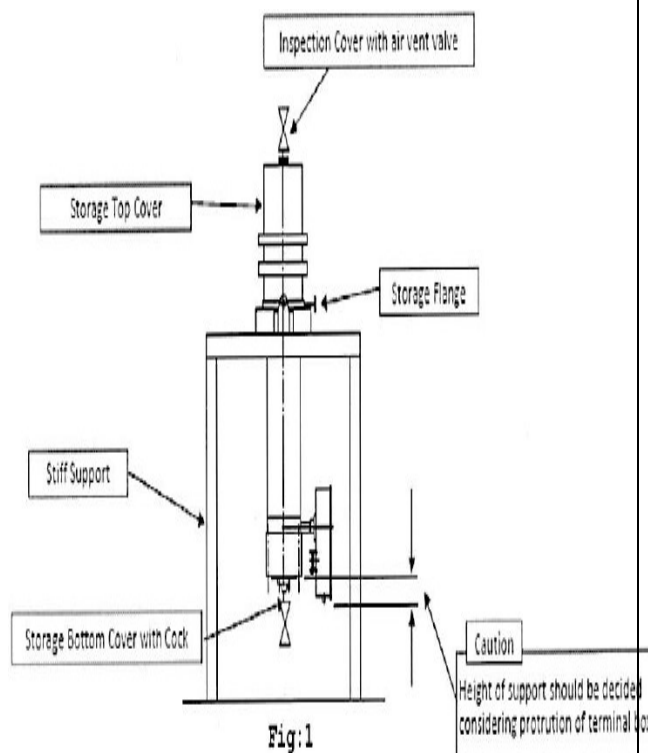
Make	Unit	MW	Pop	KV	KW
Hayward Tyler	1035 Simhadri unit 1 &2	500	8	3.3	354
Torishima	1035 Simhadri unit 3 &4	500	8	3.3	350

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

Annexure-2

Procedure for water filled storage of a spare pump motor set Hayward Tyler Pump

Motor should be stored in vertical position for water filling storage of BCW pump motor as shown in Fig:1. Care to be taken so that at no point of time it is stored in Horizontal position during water filling storage. Height of support for motor should be decided considering protrusion of the motor terminal box.



TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

Inhibitor concentration of chemically pure glycerin and distilled water in a 50/50 solution by volume, unless otherwise stated. A tolerance of $\pm 5\%$ on the normal glycerin/water ratio.

The distilled water used must have a conductivity of less than 10 reciprocal megohms (micro siemens) per centimeter.

Check the specific gravity of the inhibitor to ensure the correct concentration. The specific gravity of the mixture at 20°C is as follows: -

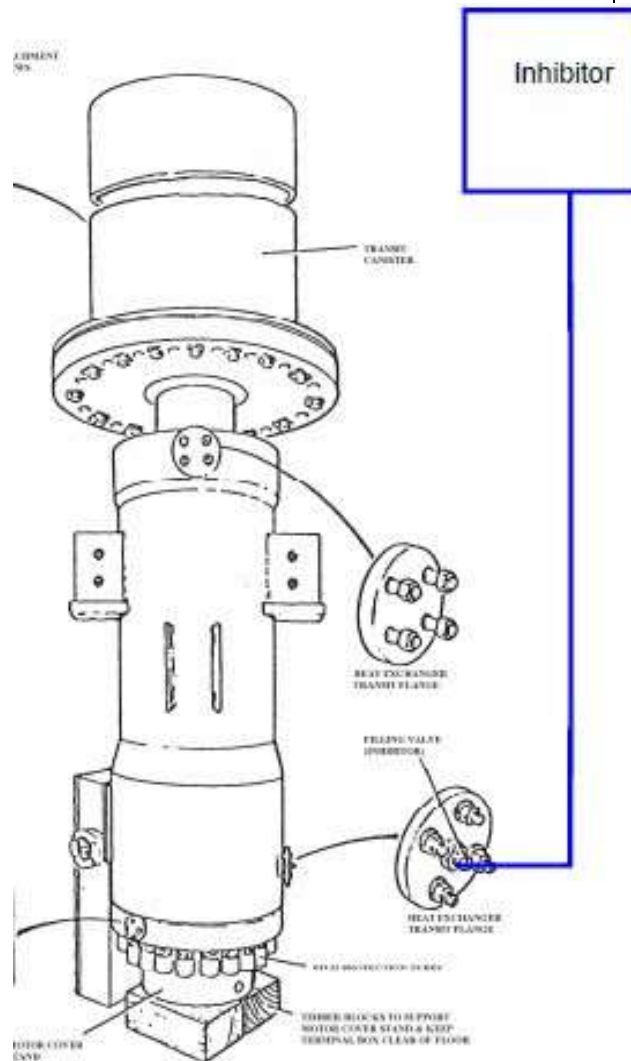
Mixture by % Volume Specific Gravity

1. 45% Glycerin/55% Water 1.11490
2. 50% Glycerin/50% Water 1.12845
3. 55% Glycerin/45% Water 1.14220
4. 60% Glycerin/40% Water 1.15605
5. 65% Glycerin/35% Water 1.16980
6. 70% Glycerin/30% Water 1.18355

This water is to be filled in BCW pump motor as shown in Fig 2.

The level of inhibitor must be above the level of

the impeller if require it must be topped up.



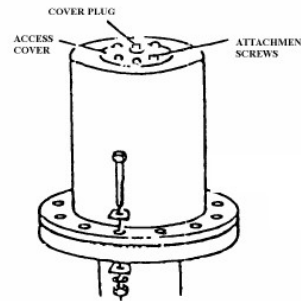
TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039,Rev:1

Annexure-3

Regular inspection in every 3 months is to be ensured as per following checklist: -

Rotation

Remove the inspection cover from the top of the transit canister. Fit impeller wrench into the impeller nut and rotate the rotating assembly seven complete revolutions in an anti-clockwise direction when viewed from the pump-end. If the shaft cannot be rotated or is very difficult to turn, the motor must be stripped down and inspected



Regular inspection in every 3 months is to be ensured as per following checklist:-

-- Check all wet preserved parts for tightness.

If leakage is detected, eliminate the cause (e.g. by further tightening screw joints or fitting new seals)

-- Check the filling level of the glycerine--water mixture

Check the volume of the glycerine--water fill; if not sufficient, replenish to the required level by adding a mixture of glycerine and water (mixing ratio 1:1).

-- Measuring the insulation resistance

Connect the measuring instrument to phase U, V or W of the cable gland and to earth (housing) and take the reading. Apply the same testing voltage as the motor voltage, or 5000 V for motors > 5 kV. The insulation resistance of the motor winding should not be less than 200 MΩ at an ambient temperature of 20°C

Preservation of motor Terminal Box:

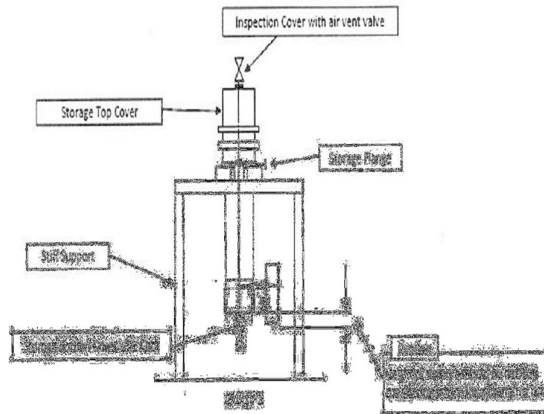
In the motor terminal box silica gel is to be kept for ensuring dryness. In the case of opening the terminal box cover, if the silica- gel color has changed to pink or is colorless, the silica-gel should be replaced by new charge before closing the terminal box cover.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039.Rev:1

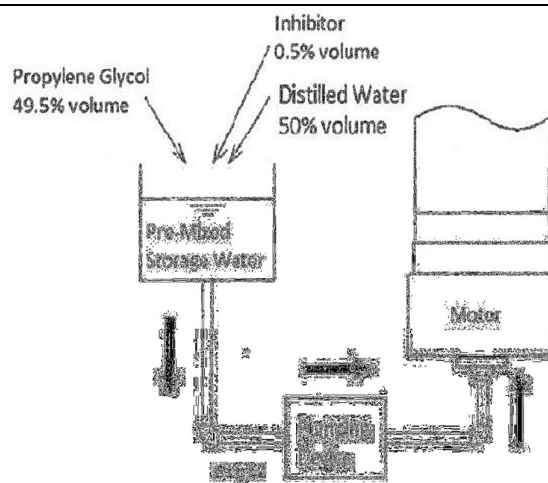
Annexure-4

Procedure for water filled storage of a spare pump motor set Torishima Pump

Motor should be stored in vertical Position for water filling storage of BCW pump motor as shown in Fig:1. Care to be taken so that at no point of time it is stored in Horizontal position during water filling storage. Height of support for motor should be decided considering protrusion of the motor terminal box.



Distilled water 50% by volume, Inhibitor (K_2HPO_4 i.e Dipotassium Phosphate) 0.5% by volume and Propylene Glycol (JIS K 2234 recommended by M/s Torishima) by 49.5% volume are to be premixed in a storage tank. The conductivity of Distilled water should be less than 10 micro siemens/cm. This water is to be filled in BCW pump motor as shown in Fig 2.



Regular inspection in every 3 months is to be ensured as per following checklist:-

-- Visual Inspection

- Checking for any leaks by visual inspection

-- Checking of water level

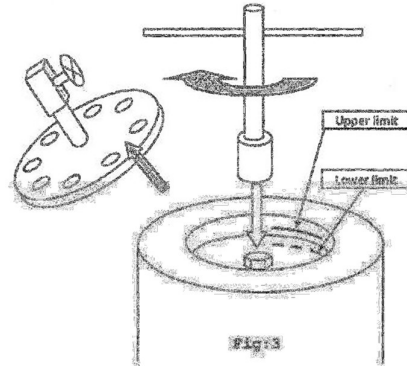
- Checking the water level after opening inspection cover

-- Measuring the insulation resistance

Connect the measuring instrument to phase U, V or W of the cable gland and to earth (housing) and take the reading. Apply the same testing voltage as the motor voltage, or 5000 V for motors > 5 kV. The insulation resistance of the motor winding should not be less than 200 MΩ at an ambient temperature of 200C

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039.Rev:1

c. Rotation checking (Ref Fig:3)
 Open inspection cover
 Set box spanner onto impeller bolt
 Rotate the rotor and observe 10 complete revolutions in clock- wise direction viewed from top. The Rotor should not be rotated in CCW (counter clock wise) direction when viewed from top to avoid damaging the lock washer for Impeller bolt.



Preservation of motor Terminal Box:

In the motor terminal box silica gel is to be kept for ensuring dryness. In the case of opening the terminal box cover, if the silica- gel color has changed to pink or is colorless, the silica-gel should be replaced by new charge before closing the terminal box cover.

Use silica-gel: PQ5up blue 5009 (Fuji Gel Sangyo Ltd.) or equivalent.

TITLE	Operation and Maintenance Guidelines for BCW Pumps
LMI No.	LMI/OGN/OPS/SYST/039.Rev:1

Annexure-5: Population of BCW pumps at NTPC Stations & JV

Make	Unit	MW	Pop	KV	KW
Hayward Tyler	1005_Rihand-I	500	6	3.3	350
	1018_Vindya-III	500	6		
	1018_Vindya-II	500	6		
	1026_Talcher-K-II	500	12		354
	1035_Simhadri-I	500	8		
	1025_Kahalgaon-II	500	9		
	1019_Sipat-II	500	6		365
	1028_Barh-II	660	2		420
	1019_Sipat-I	660	3		580
	1053_Darlipali-I	800	2		900
	1004_Singrauli-II	500	6	6.6	298
	1017_Korba-II	500	9		336
	1033_Ramagundam-II	500	9		
KSB	1005_Rihand-II	500	6	3.3	375
	1043_Mouda-II	660	2		500
	2015_NPGCL	660	3		500
	1049_Gadarwara-I	800	2		588
	1033_Ramagundam-III	500	3	6.6	350
Torishima	1005_Rihand-III	500	6	3.3	350
	1011_Dadri-C-II	500	6		
	1018_Vindya-IV	500	6		
	1018_Vindya-V	500	3		
	1006_Unchahar-II	500	3		
	1035_Simhadri-II	500	8		
	1043_Mouda-I	500	6		
	8883_Jhajjar-I	500	9		
	8884_Vallur-I	500	9		
	1028_Barh-I	660	3		420
	1054_Khargone-I	660	2		450
	1046_Kudgi-I	800	3		460
	1047_Lara-I	800	2		
	1007_Tanda-II	660	2		500
	1045_Solapur-I	660	2		550
	8882_Meja-I	660	2		
	1017_Korba-III	500	3	6.6	336
	1024_Farakka-III	500	3		380